



Society of Petroleum Engineers

SPE-227089-MS

Regional Assessment of Natural Gas Storage Deliverability in South-Central U.S. Using Stacking Machine Learning Models

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227089-MS](https://doi.org/10.2118/227089-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Summary

Underground natural gas storage (UNGS) is an important mechanism for stabilizing the supply and demand of natural gas, especially as more and more renewable energy sources are added to the power grid and the usage of natural gas varies with the seasons. In this study, a stacking machine learning framework is proposed to predict deliverability of UNGS sites in South-Central region, the United States. The stacking model integrates four base models—Random Forest, XGBoost, Support Vector Regression, and K-Nearest Neighbors—and utilizes six key factors: working gas volume, base gas volume, total storage capacity, surface temperature, geographic region, and operator type. The modeling shows excellent predictive performance, an R^2 score of 0.9979, a RMSE of approximately 23,632 Mcf, and a MAE of 6,675 Mcf. SHapley Additive exPlanations (SHAP) analysis identifies working gas as the key predictor, with secondary factors varying with field type—base gas for salt domes and total capacity for depleted fields. These findings highlight the prospects of machine learning in enhancing the planning and operational efficiency of storage systems.

Introduction

Underground natural gas storage (UNGS) facilities—encompassing depleted hydrocarbon reservoirs, saline aquifers, and salt cavern formations—serve as critical infrastructure for balancing gas supply and demand fluctuations (Kalair et al., 2021; Zhou et al., 2021). With growing integration of variable renewable energy sources into power systems, the capacity for swift gas delivery becomes essential to ensure grid stability and address surges in electricity demand. Currently, more than 700 UNGS sites exist worldwide, with the United States leading in the number of installations and overall storage capacity (Fig. 1; Al-Shafi et al., 2023). Deliverability, the maximum extraction rate of natural gas, is the key indicator for evaluating the effectiveness of UNGS systems, fundamentally determining operational responsiveness and commercial viability.

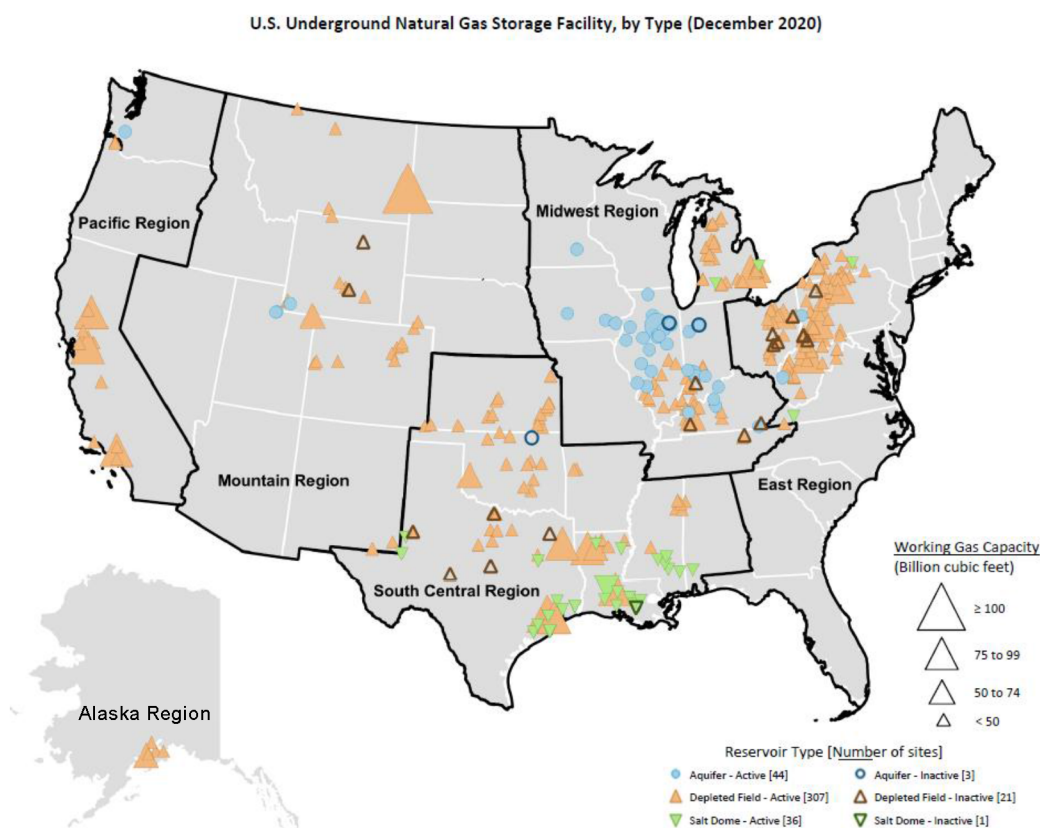


Figure 1—Distribution of underground natural gas storage in the USA (EIA, 2024a)

Traditional approaches for predicting deliverability, such as the material balance method, depend on idealized assumptions regarding reservoir homogeneity (Ali, 2021, 2022; Thanh et al., 2022a, b). However, these assumptions are not realistic under actual operating conditions. Additionally, these models require comprehensive reservoir characterization data, which are often unavailable or incomplete across most storage sites, hindering such methods implementation. In response to these limitations, data-driven machine learning (ML) methods, such as Random Forest (RF) and Support Vector Machines (SVR), have been increasingly applied to predict deliverability based on key variables such as working gas, base gas, and total field capacity. Nevertheless, many of these ML models are designed to specific regional or geological conditions and therefore may suffer from limited generalizability (Hussain et al., 2025; Wei et al., 2025; Liu et al., 2023; Mohammed and Kora, 2023; Li et al., 2022).

To improve predictive performance and applicability across diverse storage systems, this study proposes a stacking ML framework, integrating multiple algorithms—RF, XGBoost, SVR, and K-Nearest Neighbors (KNN)—to predict deliverability. In addition to traditional capacity-related features, we incorporate variables reflecting environmental and operational differences, including surface temperature, geographic location, and operator ownership type. The model's interpretability is enhanced using SHAP (SHapley Additive exPlanations), which quantifies the individual impact of each feature on the predicted output. Our analysis focuses on the high-capacity South-Central U.S. region, aims to support data-driven operational planning and contribute to more flexible, resilient energy infrastructure.

Data and Methodology

This study utilizes publicly available data from the U.S. Energy Information Administration (EIA) and other sources spanning 2014–2024 to evaluate UNGS deliverability (EIA, 2024a, b; Fig. 1). The dataset

includes field-level monthly statistics, which were aggregated and cleaned to form a structured input matrix for machine learning applications.

Input Variables

Six key features were selected based on relevance and data availability: (1) Working gas capacity: indicates the volume of gas that can be injected and withdrawn under normal operation. (2) Base gas capacity: reflects the permanent gas volume retained to maintain reservoir pressure. (3) Total field capacity: sum of working and base gas capacities, representing overall storage potential. (4) Surface temperature: a proxy for seasonal and regional environmental variations influencing gas behavior. (5) Ownership type: categorized as interstate pipeline companies, local distribution firms, or independent operators, representing operational and regulatory differences. (6) Location (region code): captures geological and infrastructure-related spatial differences. These parameters were selected to reflect both physical and organizational attributes affecting gas deliverability. Features were normalized before model training.

Stacking Model Framework

To enhance prediction performance and generalizability, a stacking ensemble learning strategy was adopted. The base learners include: RF, XGBoost, SVR, and KNN. Each base model was trained using 5-fold cross-validation. Their predictions were then passed to a meta-learner, which produced the final deliverability prediction. This two-level framework integrates the strengths of different algorithms, improving robustness (Wei et al., 2024). Model performance was evaluated using three standard regression metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Coefficient of Determination (R^2).

Sensitivity Analysis

To interpret model predictions, SHAP was employed to reveal the marginal impact of each feature on the predicted deliverability, allowing for intuitive understanding of feature importance and nonlinear interactions.

Results

Deliverability Distribution

Deliverability by State in South-Central. Figure 2 illustrates the distribution of maximum daily gas deliverability in the South-Central region across various U.S. states, including Arkansas (AR), Kansas (KS), Louisiana (LA), Mississippi (MS), Oklahoma (OK), Texas (TX), and Alabama (AL). This area accounts for a total of 7084 observations, representing a wide range of UNGS facilities. Notably, Alabama demonstrates the highest median deliverability, surpassing 2.5×10^6 Mcf, even though it has relatively fewer records ($n = 211$). In contrast, states like Louisiana and Texas display greater variability, reflected by broader interquartile ranges and more outliers. This indicates a wide range of storage capacities and operational setups, likely resulting from the presence of multiple storage types and facility sizes. Arkansas and Kansas, on the other hand, show the lowest median deliverability, which may reflect limited infrastructure capacity or more conservative gas withdrawal practices.

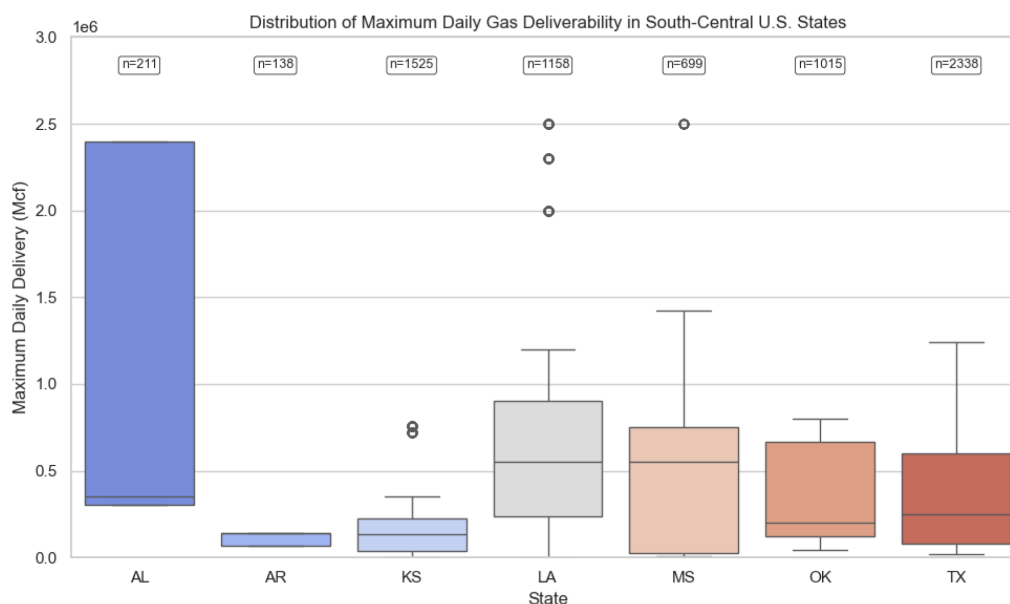


Figure 2—Boxplot showing the distribution of maximum daily gas deliverability across selected South-Central U.S. states. The y-axis represents the daily maximum output of underground natural gas storage facilities (in Mcf), and the x-axis lists individual states. Sample sizes (n) are annotated above each box.

Deliverability by Field Type. Figure 3 compares the maximum daily deliverability of underground gas storage facilities between two major field types: depleted fields and salt domes. The results indicate clear distinctions in both central tendency and variability. Salt dome facilities exhibit significantly higher median deliverability and a broader interquartile range, with some outliers reaching up to 2.5×10^6 Mcf. This performance reflects their favorable cavern geometry, higher injection/withdrawal efficiency, and lower cushion gas requirements. In contrast, depleted fields show lower median values and a narrower distribution, indicating more constrained withdrawal capacity likely due to reservoir heterogeneity and aging infrastructure. With over 5400 observations, depleted fields remain the dominant storage type in terms of quantity, but salt domes contribute disproportionately to peak deliverability capacity.

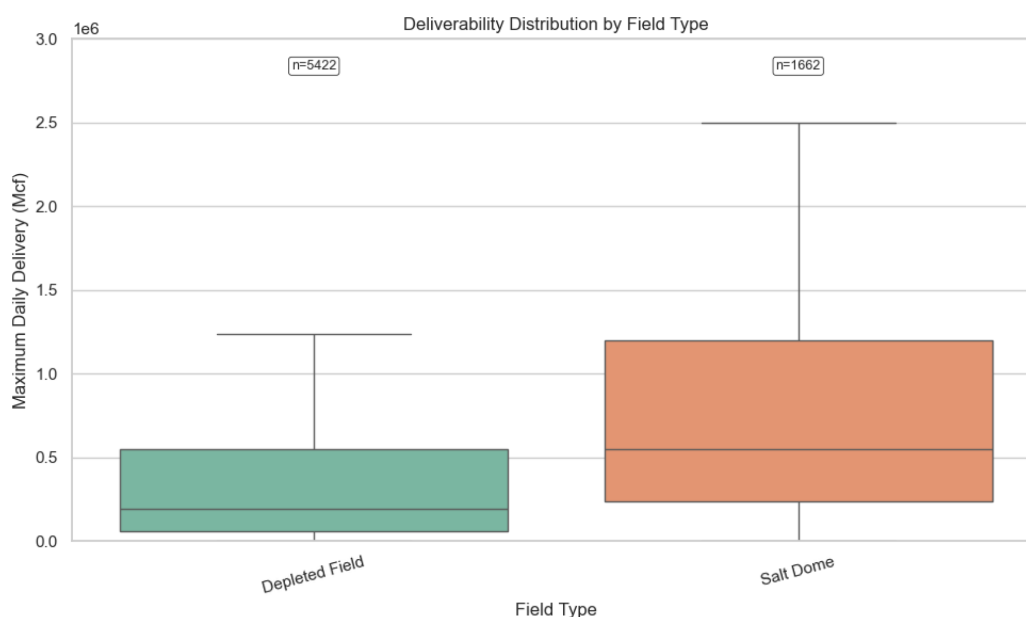


Figure 3—Boxplot showing the distribution of maximum daily deliverability (Mcf) by underground storage field type.

Deliverability by Company Ownership Type. Figure 4 presents the distribution of maximum daily deliverability across five categories of company ownership: storage operators, interstate companies, intrastate companies, municipally owned entities, and gatherers. Storage operators and interstate companies exhibit the broadest distributions and the highest number of high-deliverability outliers, with median values exceeding 0.4×10^6 Mcf. This suggests their significant role in large-scale, flexible storage operations, often serving as system-wide balancers. Intrastate companies also display moderate deliverability ranges, though with slightly lower medians and fewer extreme outliers. In contrast, municipally owned and gatherer-type operators show both limited deliverability and narrow variance, likely reflecting localized, low-throughput systems with limited infrastructure scale. These distinctions underscore the operational and strategic differences driven by company ownership structure, with independent storage operators particularly excelling in high-capacity deliverability.

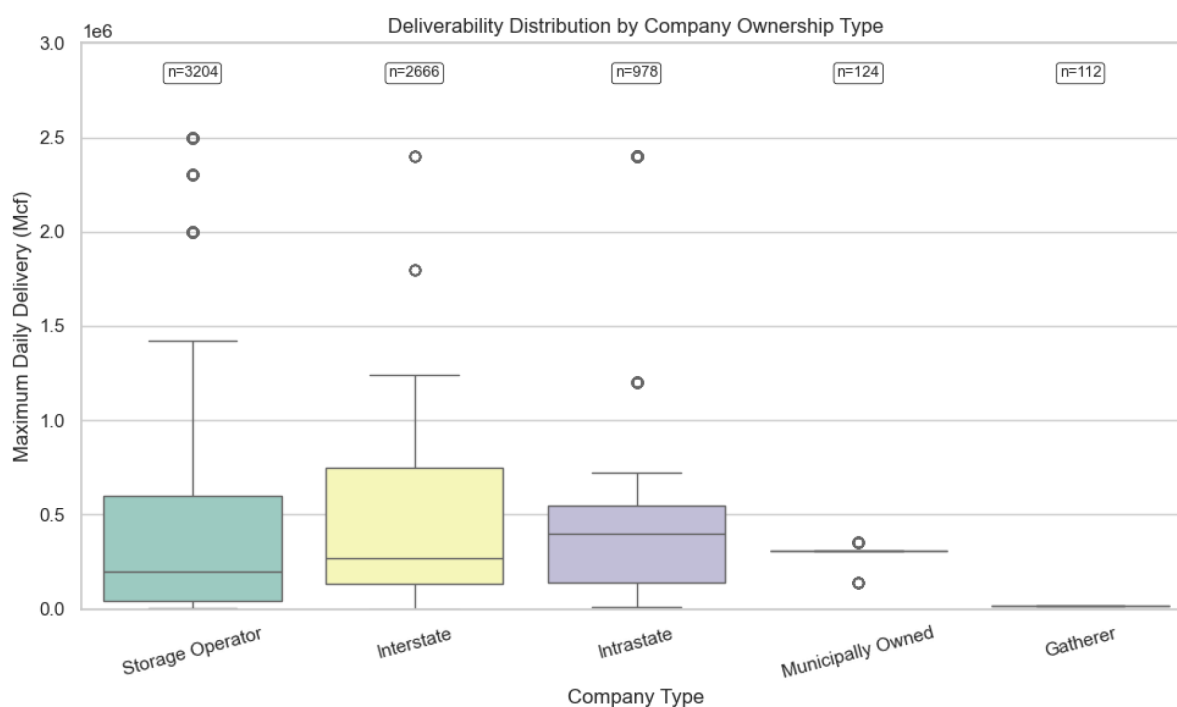


Figure 4—Boxplot illustrating the distribution of maximum daily gas deliverability (Mcf) by company ownership type.

Deliverability by Surface Temperature. Figure 5 presents the distribution of maximum daily deliverability across five surface temperature ranges. A general trend of increasing deliverability variability with higher temperature ranges is observed. Sites located in areas with surface temperatures exceeding 50°F —particularly within the $50\text{--}65^{\circ}\text{F}$ and $65\text{--}80^{\circ}\text{F}$ bins—exhibit broader interquartile ranges and higher median deliverability. This may reflect the geographic overlap between warmer regions and large-capacity storage facilities, particularly salt caverns in the South-Central U.S. By contrast, sites in colder regions ($<35^{\circ}\text{F}$) display significantly lower median deliverability and narrower spread, which may be due to limited infrastructure capacity or more conservative operational strategies during peak heating demand. Interestingly, while median deliverability values continue to rise slightly in the $>80^{\circ}\text{F}$ range, the overall spread remains broad, indicating that regional or operational characteristics—rather than temperature alone—play a major role in shaping performance.

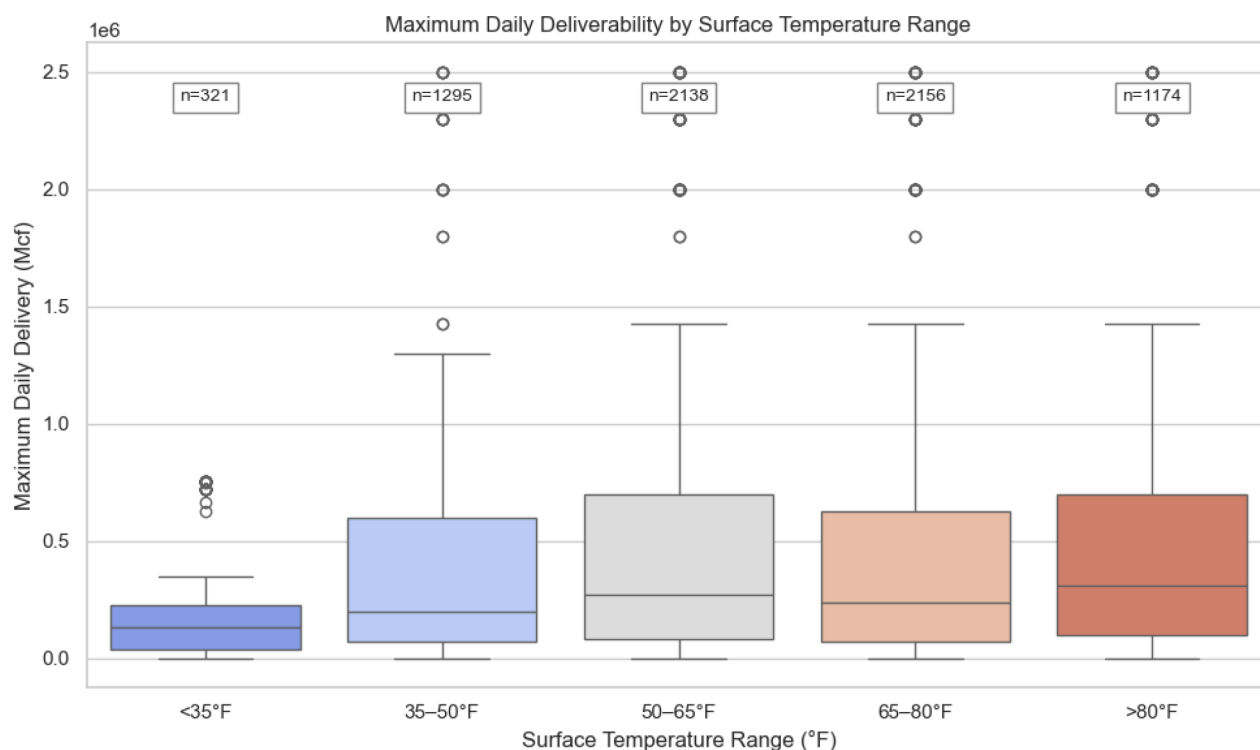


Figure 5—Boxplot showing maximum daily gas deliverability (Mcf) across surface temperature ranges.

Evaluation of Stacking Model Performance

Performance Evaluation of Models. The predictive performance of the stacking ensemble model was evaluated using the test dataset. The results demonstrate excellent agreement between predicted and actual deliverability values, with an R^2 score of 0.9979, a RMSE of approximately 23,632 Mcf, and a MAE of 6,675 Mcf. These metrics indicate both high overall accuracy and minimal deviation from true values.

As shown in Figure 6, the scatter plot of actual versus predicted deliverability values reveals a near-perfect alignment along the 1:1 diagonal, confirming the model's ability to generalize across a wide deliverability range (up to 2.5×10^6 Mcf). Figure 7 further illustrates the distribution of residuals (i.e., prediction errors), which follow a sharp, symmetric peak centered near zero. This suggests unbiased predictions with low variance, and no evidence of systemic under- or overestimation. Collectively, the evaluation confirms that the stacking model achieves robust and reliable performance for deliverability prediction across heterogeneous geological and operational conditions.

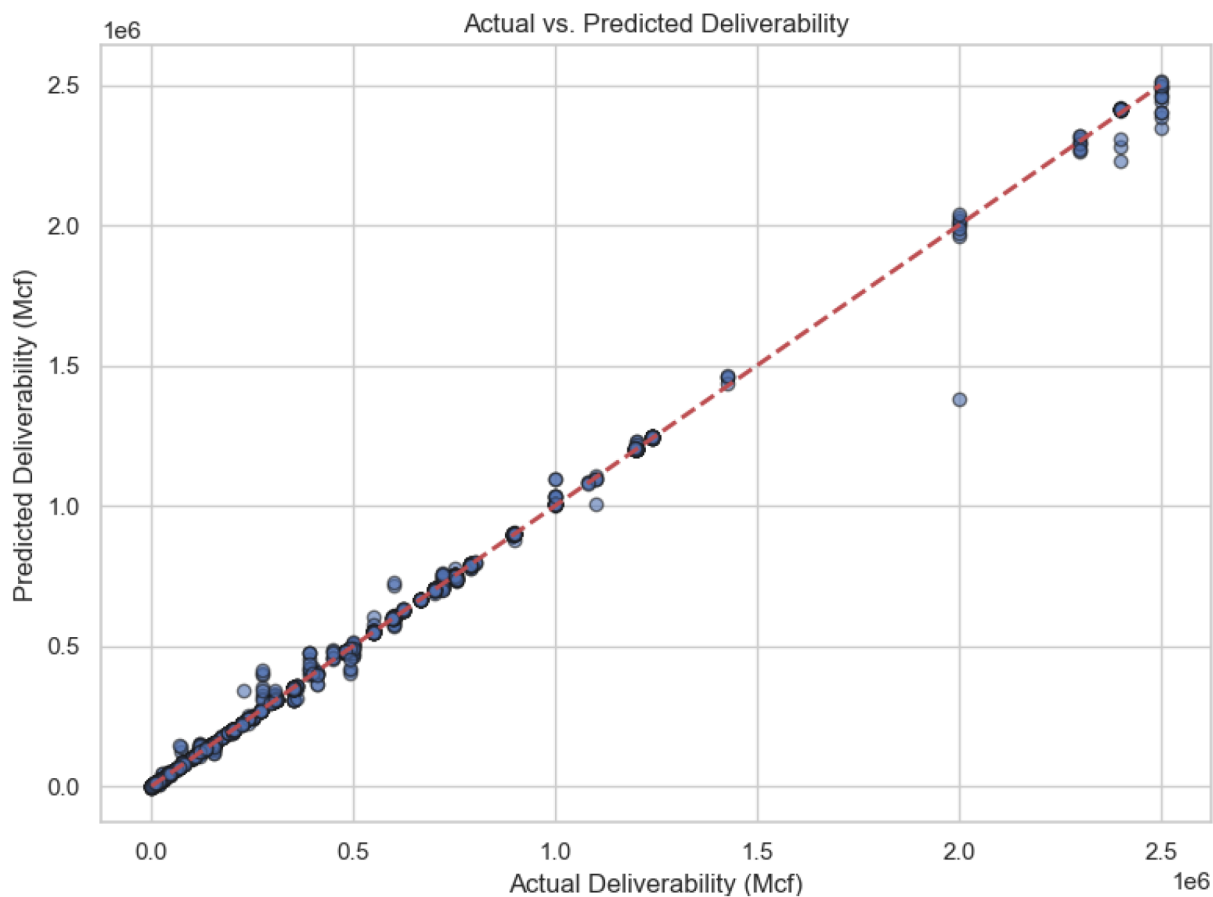


Figure 6—Scatter plot comparing actual and predicted maximum daily gas deliverability (Mcf) using the stacking ensemble model. The red dashed line indicates the 1:1 line of perfect agreement. The tight clustering along this line reflects the model's high prediction accuracy ($R^2 = 0.9979$).

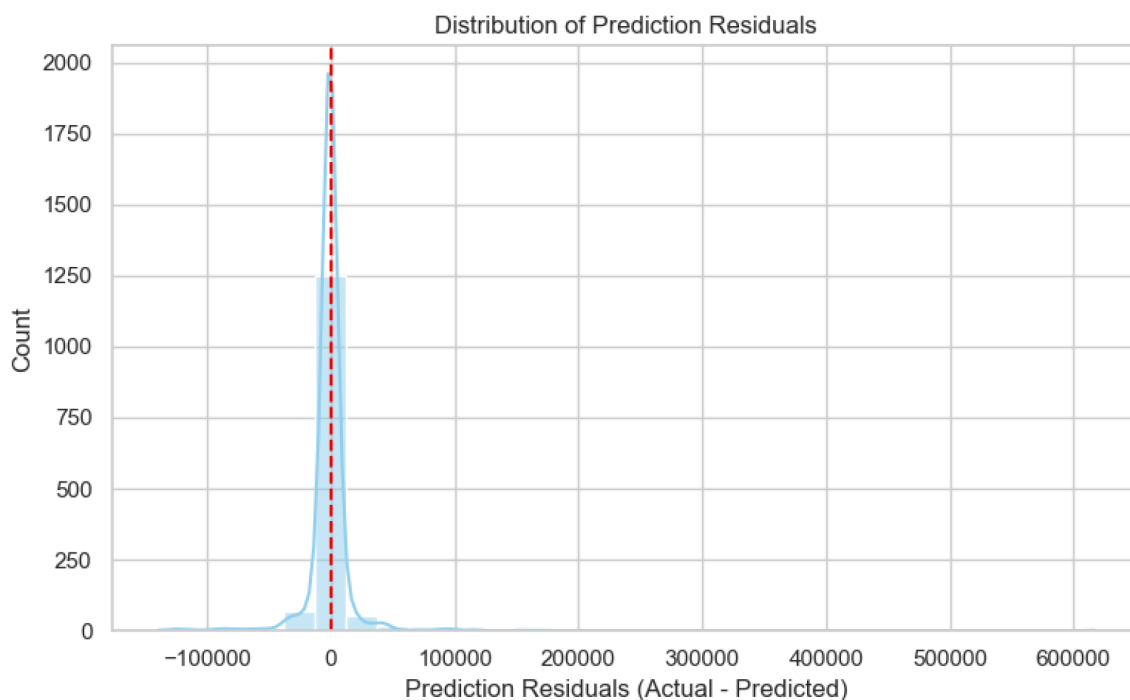


Figure 7—Histogram showing the distribution of prediction residuals (actual minus predicted deliverability). The narrow and symmetric shape, centered around zero, indicates minimal bias and low prediction error variance in the stacking model.

Sensitivity Analysis

To better understand the behavior of the stacking ensemble model, SHAP (SHapley Additive exPlanations) analysis was conducted using the numerical input features (Fig. 8). The overall SHAP summary plot (Fig. 8a) reveals that Working Gas Capacity is the most impactful feature in predicting deliverability. Higher values of this variable are strongly linked to greater output predictions, which aligns with engineering expectations—since working gas represents the active volume available for injection and withdrawal cycles, it directly governs a facility’s peak supply potential. Base Gas and Total Field Capacity also show meaningful contributions. While base gas is not recoverable, it maintains reservoir pressure and structural integrity, and thus supports sustained withdrawal rates. Facilities with larger total capacities often reflect better-developed infrastructure, contributing to stable and higher deliverability. Surface Temperature, although less influential overall, shows subtle effects that may correspond to seasonal demand variation or temperature-sensitive operational practices in specific regions.

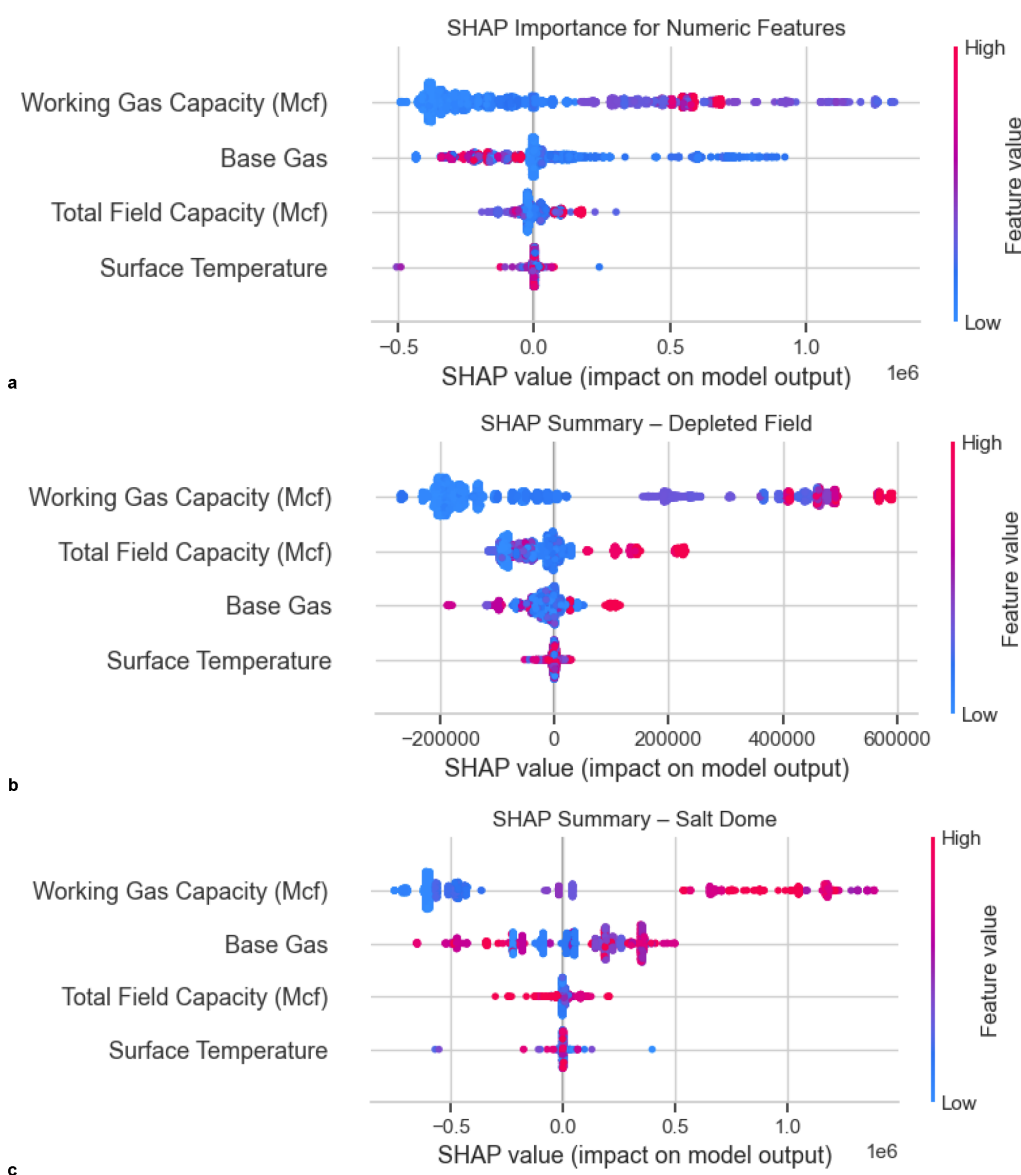


Figure 8—SHAP summary plot showing the impact of numeric features on the predicted maximum daily deliverability for (a) all field types; (b) depleted field, (c) salt dome.

To further investigate geological influences, SHAP analysis was separately conducted for depleted fields and salt dome storage types (Figs. 8b and 8c). Results reveal distinct patterns across the two storage classes: In depleted fields (Fig. 8b), working gas still dominates, but its SHAP values are more modest in scale. This suggests that in such formations—typically characterized by heterogeneous rock properties and pressure depletion—other factors like infrastructure condition or residual pressure may limit the impact of working gas alone. In contrast, salt dome facilities (Fig. 8c) exhibit much larger SHAP value magnitudes, particularly for working gas and total field capacity. This reflects the highly efficient withdrawal behavior of salt caverns, where high deliverability is supported by steep pressure gradients and low cushion gas requirements. Base gas also shows stronger effects in this group, indicating its role in supporting rapid cycling performance.

Discussion

The stacking ensemble model presented in this study achieved outstanding predictive performance across a heterogeneous set of U.S. UNGS facilities, supported by both quantitative metrics ($R^2 = 0.9979$) and residual analysis (Figs. 6, 7). The integration of four diverse base learners—RF, XGBoost, SVR, and KNN—enabled the capture of complex nonlinearities and localized trends within the dataset, while the linear meta-learner effectively mitigated overfitting by assigning adaptive weights.

The SHAP-based sensitivity analysis confirms that the stacking model effectively captures the key physical drivers behind deliverability behavior. Working gas capacity consistently emerged as the most influential variable across all storage types, reflecting its role as the core volume available for injection and withdrawal. However, the magnitude of its influence varied notably by storage type: in salt dome systems, SHAP values for working gas were substantially higher, indicating strong model sensitivity to this feature under conditions of high-frequency cycling and steep pressure gradients. In contrast, depleted fields exhibited a more moderate response, likely due to geological heterogeneity and reduced formation pressure, which diminish the marginal effect of additional working gas. Base gas and total field capacity also significantly influenced predictions in both field types. These features help maintain reservoir pressure and infrastructure scale, which affect long-term gas flow sustainability. Base gas had a greater impact in salt caverns than in depleted fields, consistent with its role in stabilizing high-deliverability systems. While surface temperature shows the weakest impact, its effect became more apparent under specific geological and operational settings.

While the model demonstrates strong predictive capabilities, it has several limitations. The model is trained on monthly averaged operational data, lacking geological or engineering parameters such as formation permeability, pressure gradients, or injection rates. This data gap may limit its effectiveness for real-time decision-making or detailed operational optimization. Nevertheless, this work illustrates that stacking ML models combined with SHAP, offer a scalable and effective approach to predicting UNGS deliverability. It provides not only accurate predictions but also meaningful operational insights.

Conclusion

This research proposes an effective stacking ML approach to predict UNGS deliverability across the South-Central U.S. region. By combining four different ML algorithms as base models and utilizing a meta-learner for final prediction, the model captures complex nonlinear dependencies and reflects operational differences among regions. SHAP analysis revealed that working gas capacity is the most critical factor influencing deliverability across all storage types. Base gas and total field capacity, are secondary influences for salt caverns and depleted fields, respectively.

The model demonstrates high predictive accuracy ($R^2 = 0.9979$) across various field types, operator categories, and climatic zones. It offers practical insights for storage optimization and infrastructure planning, especially in regions with high renewable energy penetration and seasonal gas demand

fluctuations. Despite relying on static and aggregated data, the model provides a scalable and explainable tool for strategic assessment of gas storage systems.

Future work should explore dynamic data integration, hybrid physics-informed frameworks, and regional calibration to further enhance model granularity and transferability. Ultimately, this study contributes to the development of data-driven approaches for resilient, adaptive, and efficient gas storage system management in an evolving energy landscape.

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